

CS 6180 10/20

HW3

1.5 weeks

Release HW3 tonight

(Continuation of HW2
part on transformers)

on email (or
hopefully on
canvas)

www.gradescope.com

login with Northeastern account

Variational Auto **Encoders**

(generating images)

↓
pretty good at generating
images of faces of people that
don't even exist.

Encoder

Decoder

images
of
faces
of people

important
features
in the
the images

generate some
new images of
faces

Gonna generate an image x

$p(x)$

relevant features in faces:

- * eye color
- * eye shape
- * nose shape
- * shape of the face
- * eyebrows (color, shape)
- * hair

$p(x)$ ← more complex eye color

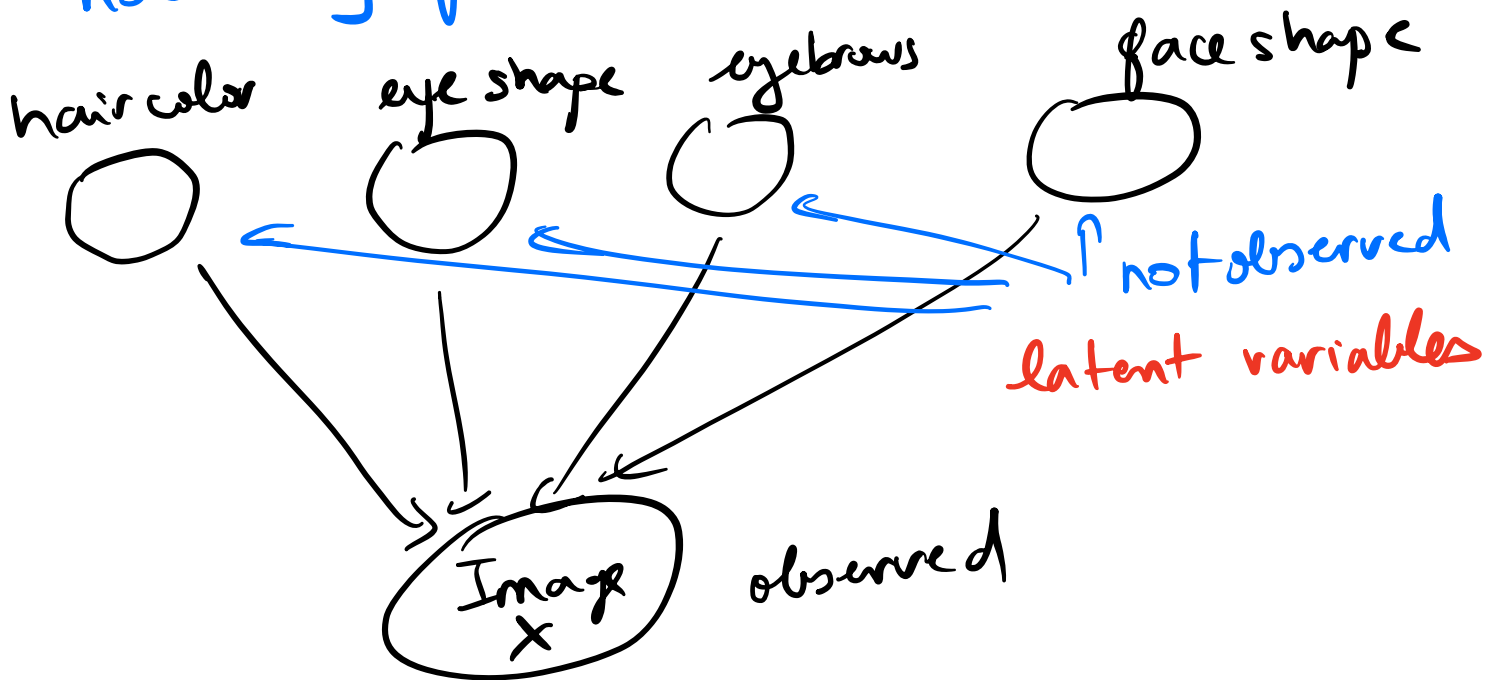
$p(x | z = \text{brown})$

↑
simpler model to learn compared to the first one

$p(x | z_1)$ ↑ eye color
 $p(x | z_2)$ ↑ hair color
 $\dots p(x | z_k)$ ↑ eyebrows

$$p(x) = p(z_1) \cdot p(x|z_1) + p(z_2) p(x|z_2) + \dots + p(z_k) p(x|z_k)$$

how many features are relevant?



0.01	0.27
0.27	0.01

image with 4 pixels

0.9	0.7
0.7	0.9

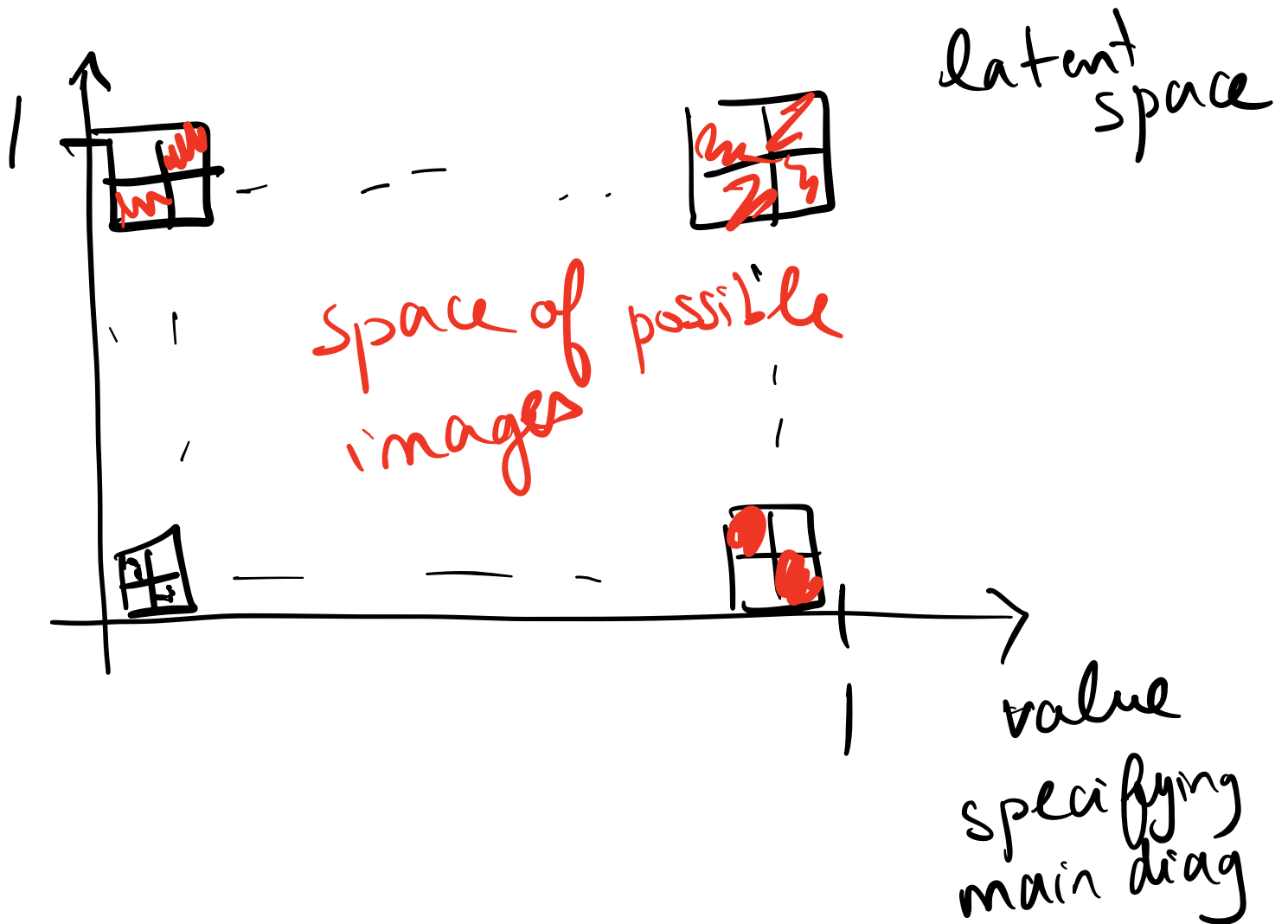
0.78	0.92
0.92	0.78

0.61	0.5
0.5	0.61

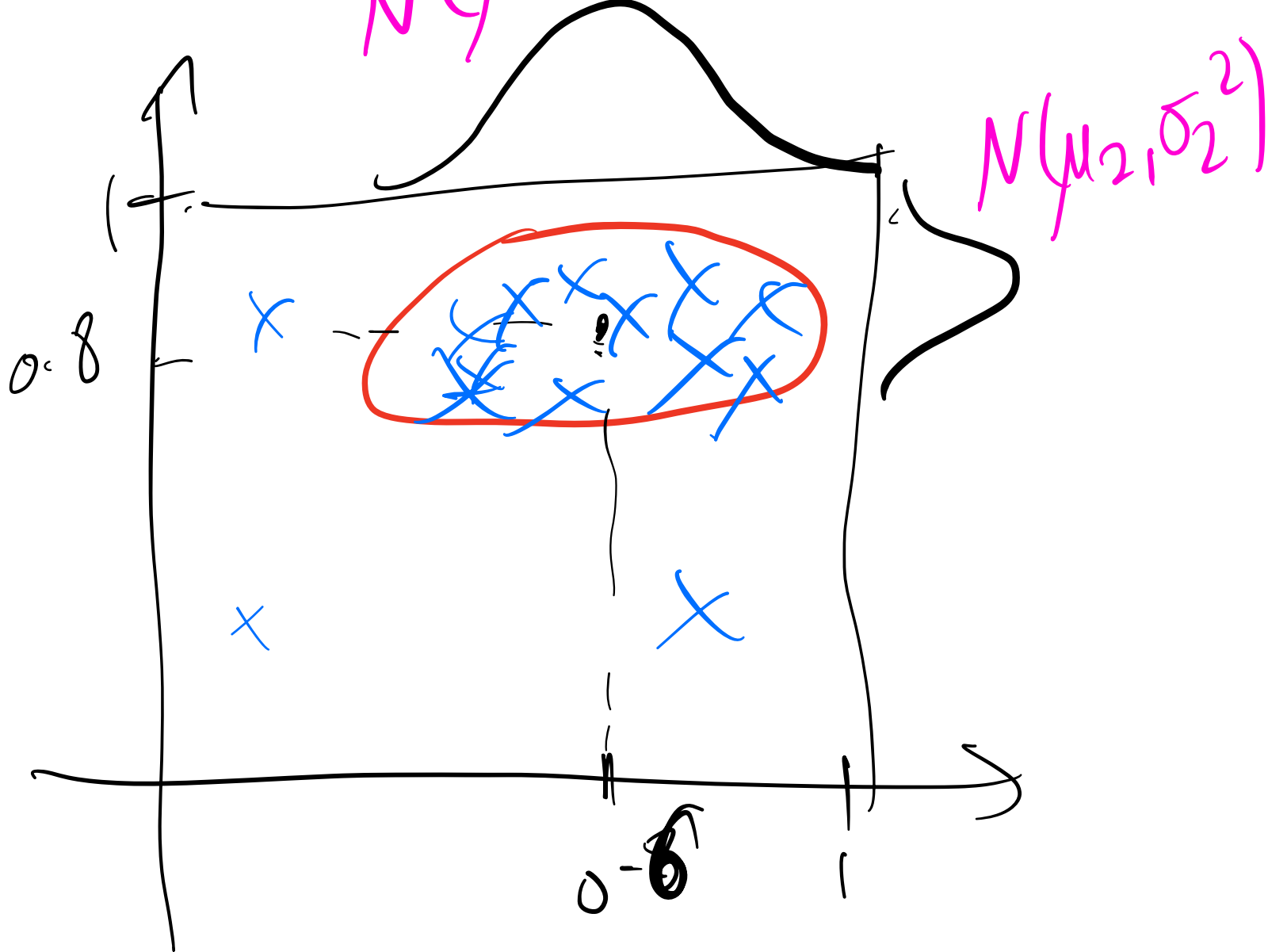
0.5	0.01
0.01	0.5

values along both diagonals are the same

⇒ instead of learning 4 values, we can instead learn 2 values.



$$r(M, \sigma^2)$$



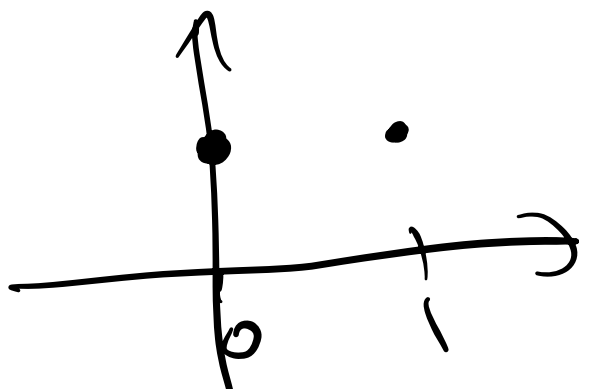
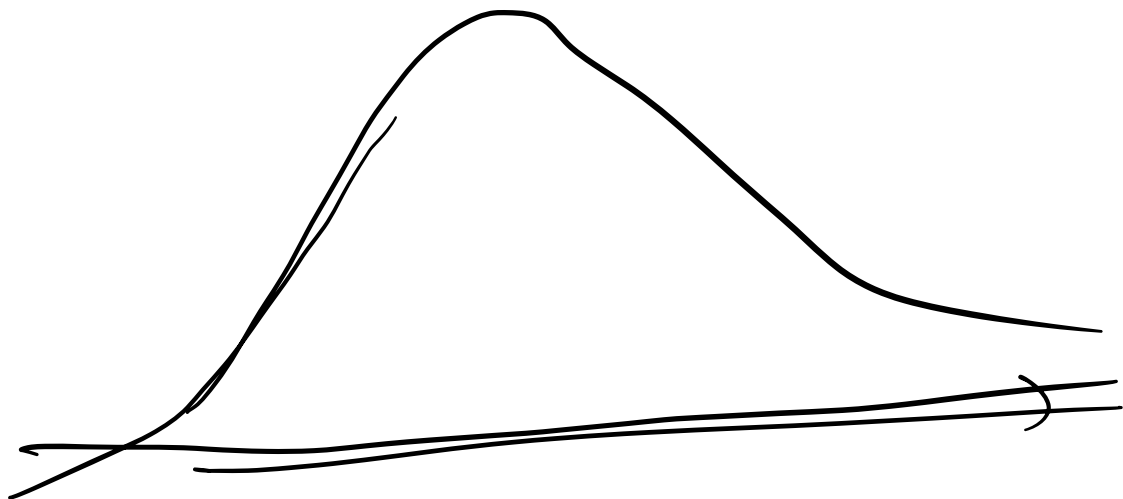
many data points
could fall in the
region in red

⇒ need to figure

out a proper
distribution in each
coordinate.

Background on normal distributions

sum up a random variables
many many times



$X_1 \sim \text{Bernoulli distribution}$

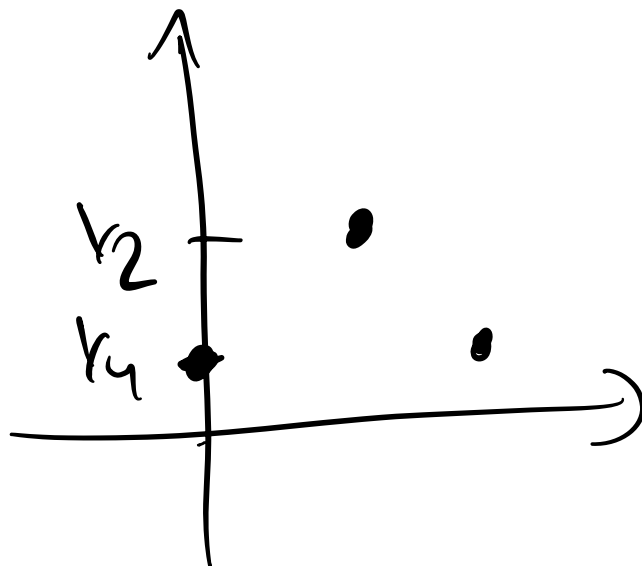
$$P(X_1) = \begin{cases} 1/2 & \text{if } X_1 = 0 \text{ (tails)} \\ 1/2 & \text{if } X_1 = 1 \text{ (heads)} \end{cases}$$

$X_1 \sim \text{Bernoulli}, X_2 \sim \text{Bernoulli}$

$$\underline{X_1 + X_2}$$

$$Y = X_1 + X_2$$

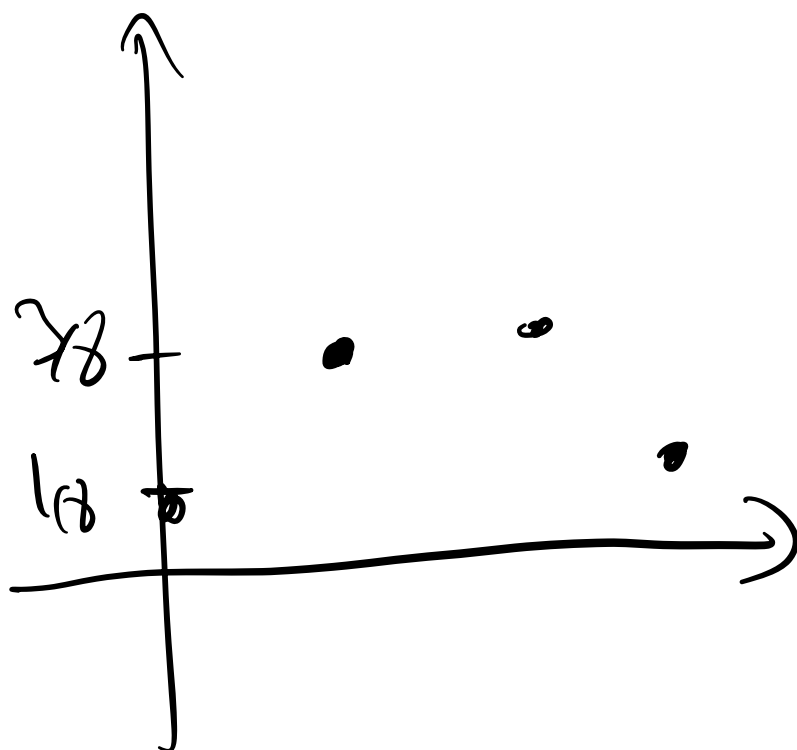
$$P(Y) = \begin{cases} 1/4 & \text{if } Y = 2 \\ 1/2 & \text{if } Y = 1 \\ 1/4 & Y = 0 \end{cases}$$



$$Y = X_1 + X_2 + X_3$$

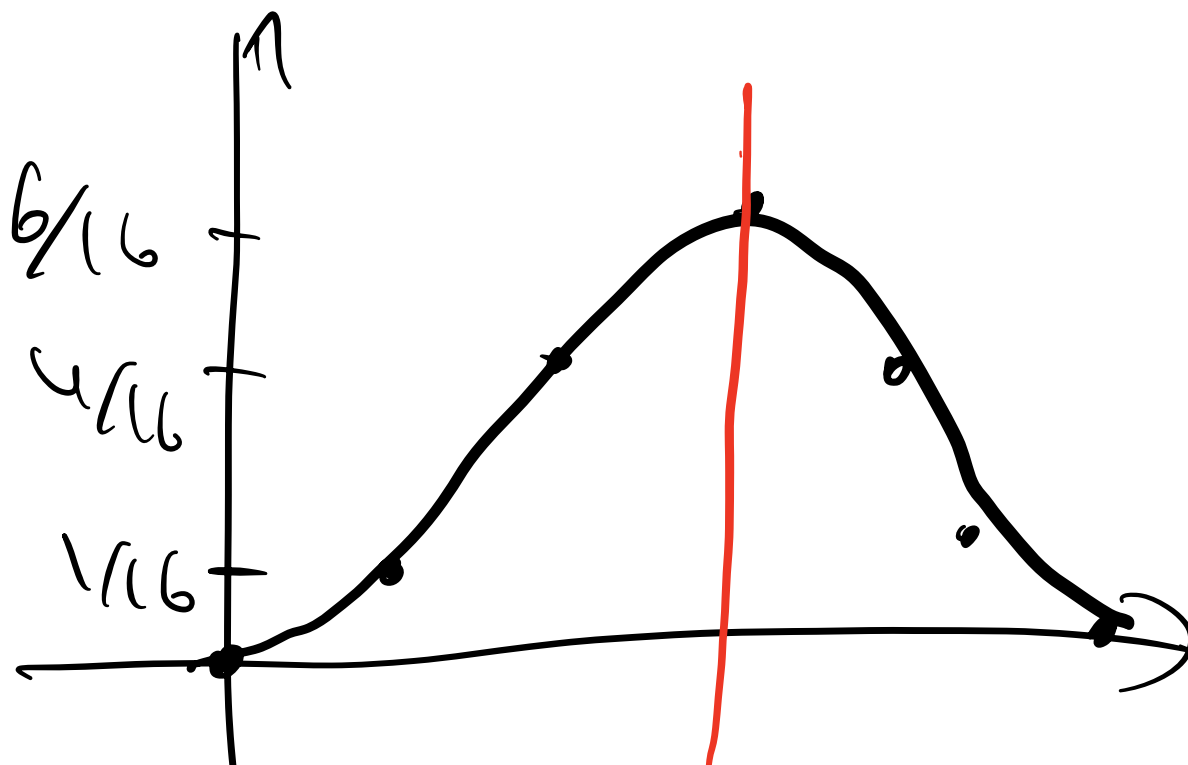
$$P(Y) = \begin{cases} 1/8 & Y=3 \\ 3/8 & Y=2 \\ 3/8 & Y=1 \\ 1/8 & Y=0 \end{cases}$$

H H T
 H T H
 T H H



$$Y = X_1 + X_2 + X_3 + X_4$$

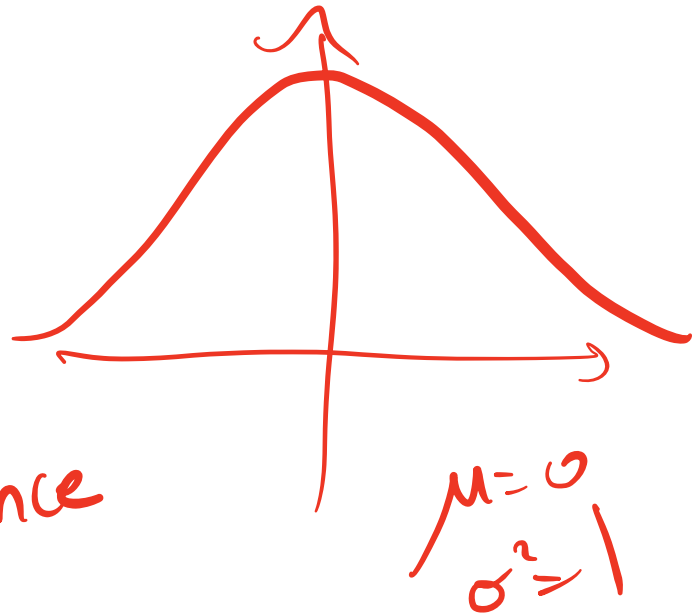
$$P(Y) = \begin{array}{ll} 1/16 & Y=4 \\ 4/16 & Y=3 \\ 6/16 & Y=2 \\ 4/16 & Y=1 \\ 1/16 & Y=0 \end{array}$$



central limit theorem

$N(\mu, \sigma^2)$
↓
mean

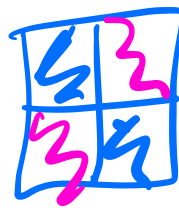
↓
variance



Back to Variational Auto encoders :

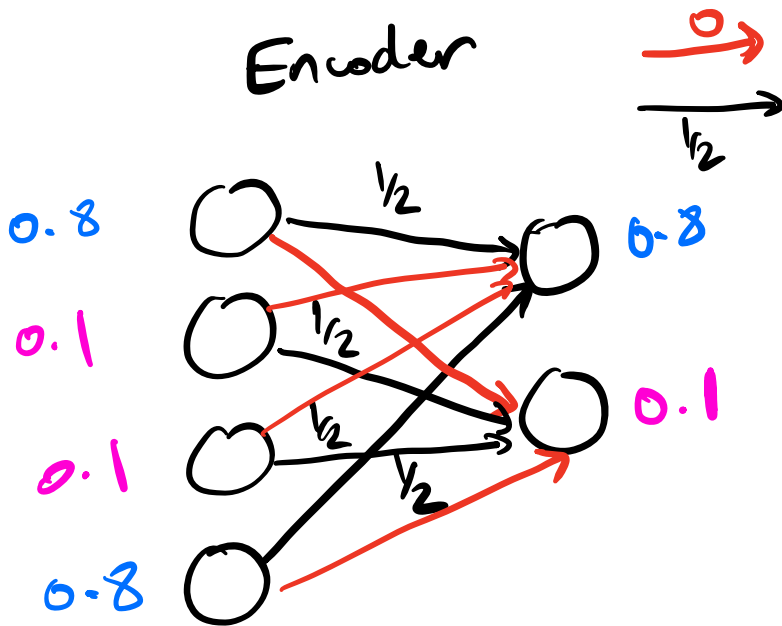
We need to figure out the means μ_1, μ_2 and variances σ_1^2 and σ_2^2 to properly train the model and generate images from the region where most of the data points come.

Goal : Generate Images

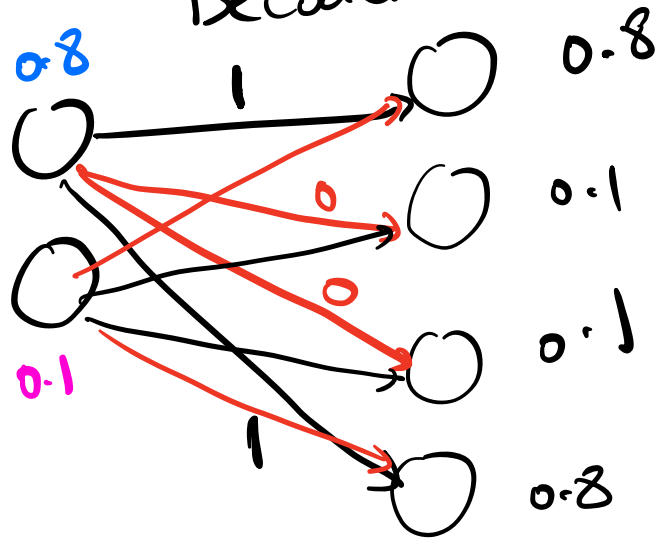


$\frac{4}{2} = 0.8$
 $\frac{3}{4} = 0.1$

Encoder



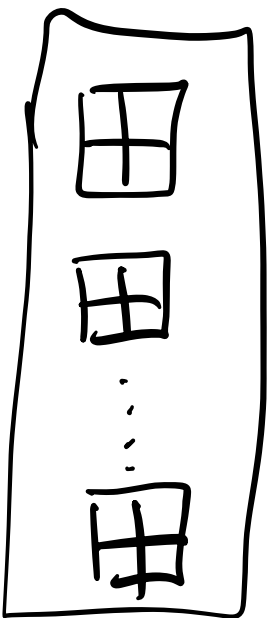
Decoder



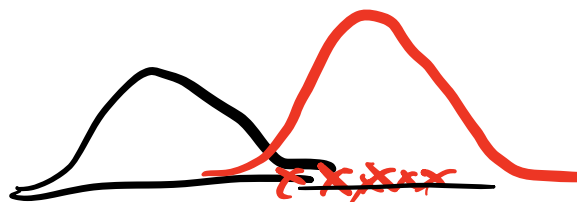
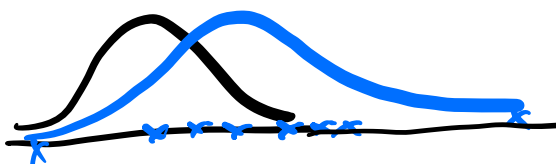
Linear encoders

$\Rightarrow$ add non linearities to be able
 to represent more complex images

we can add some sigmoids to get non linearities
 (activation functions)



Encoder



Need some metric that allows to compare two probability distributions.

p and q are two proba distributions

KL divergence

$$KL(p, q) = \sum_i p_i \log\left(\frac{p_i}{q_i}\right)$$



$$0.1 \log\left(\frac{0.1}{0.15}\right) + 0.2 \log\left(\frac{0.2}{0.15}\right) + \dots + 0.1 \log\left(\frac{0.1}{0.1}\right)$$

Background on KL divergence

Information theory

$p(x)$: proba distribution

entropy of a distribution (how much chaos, how much surprise we have in a distribution)

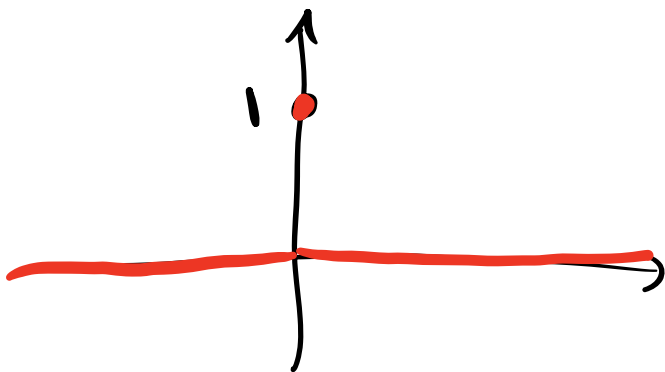
common events ($p(x)$ high)
 $\Rightarrow$ not gonna get a lot of new info
content

rare events ($p(x)$ is low)

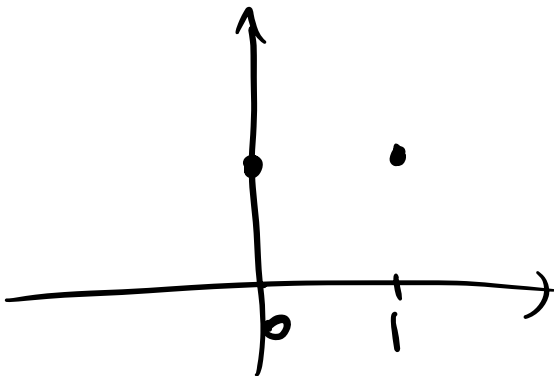
$\Rightarrow$ giving me quite a lot of new info that
I didn't know before.

$$\text{entropy}(p) = \sum_x p(x) (-\log(p(x)))$$

$$= - \sum_x p(x) \log(p(x)) = -E[\log p(x)]$$



$$\begin{aligned}\text{entropy}(p) &= - \sum_x p(x) \log(p(x)) \\ &= -1 \log(p(0)) \\ &= -1 \log(1) = 0.\end{aligned}$$



base e

$$\begin{aligned}\text{entropy}(p) &= -\frac{1}{2} \log\left(\frac{1}{2}\right) - \frac{1}{2} \log\left(\frac{1}{2}\right) \\ &= -\log\left(\frac{1}{2}\right) = \log(2)\end{aligned}$$

Normal distribution has
a huge entropy
(distribution with
the largest
entropy)

* Cross entropy

- true distribution of data $P(x)$
- model that we are currently using to
get some idea of surprise (entropy)
 $Q(x)$

$$CE(p, q) = - \sum_x \underline{p(x)} \log \underline{Q(x)}$$

average amount of surprise we are
seeing when using $Q(x)$ instead
of the actual true distribution
 $p(x)$

intuition:

$p_{\text{data}}(x)$

$P_{\theta}(x)$

unknown

model
distribution

how much

new
info / surprise /
entropy
in the data

$$- \sum_x p_{\text{data}}(x) \log p_0(x)$$

$$KL(p, q) = \underbrace{CE(p, q) - \text{entropy}(p)}$$

metric to
assess how
similar p and
 q are to
each
other.

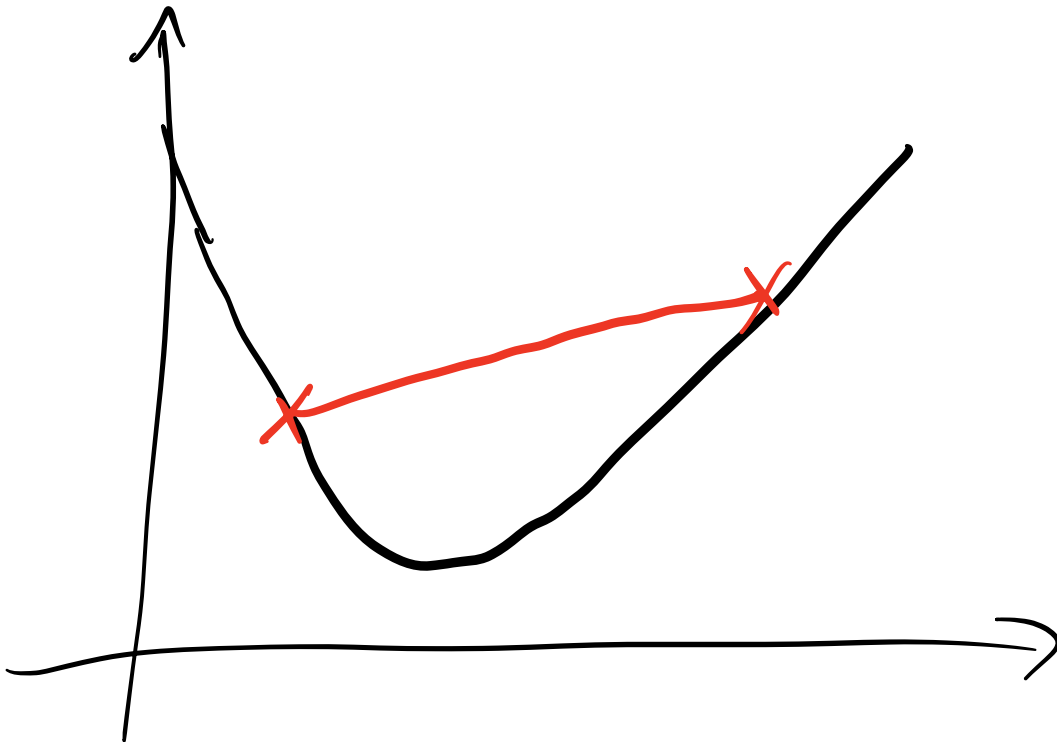
notice that if

$$KL(p, q) = 0 \Rightarrow \underline{CE(p, q)} = \underline{\text{entropy}(p)}$$

Want $KL(p, q) \geq 0$ for
any two distributions

to represent some distant metric
between two distributions

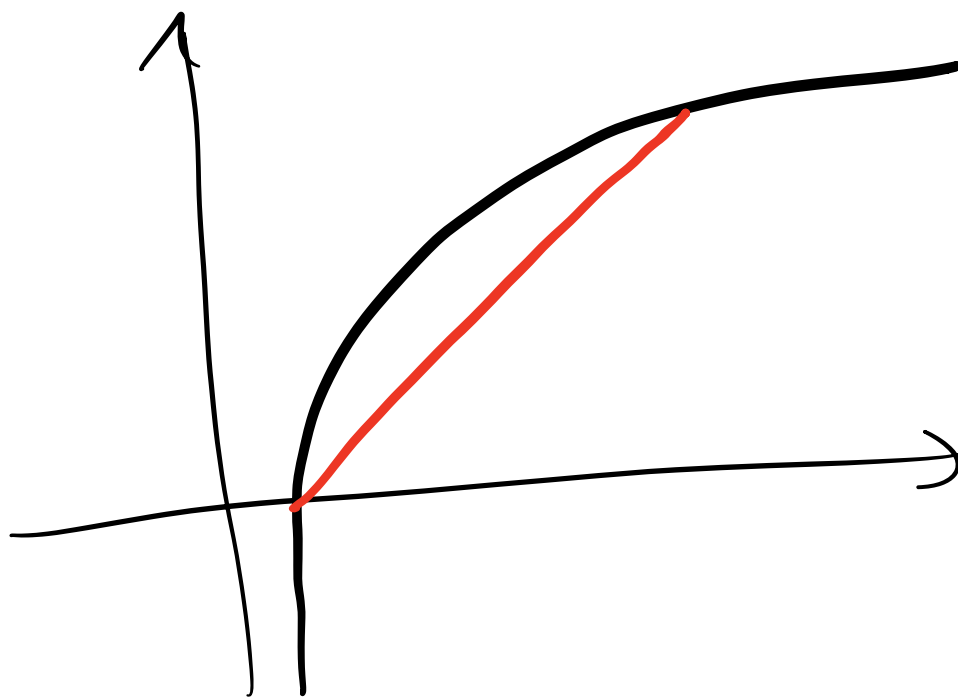
Jensen's inequality



always true for any f that
is convex

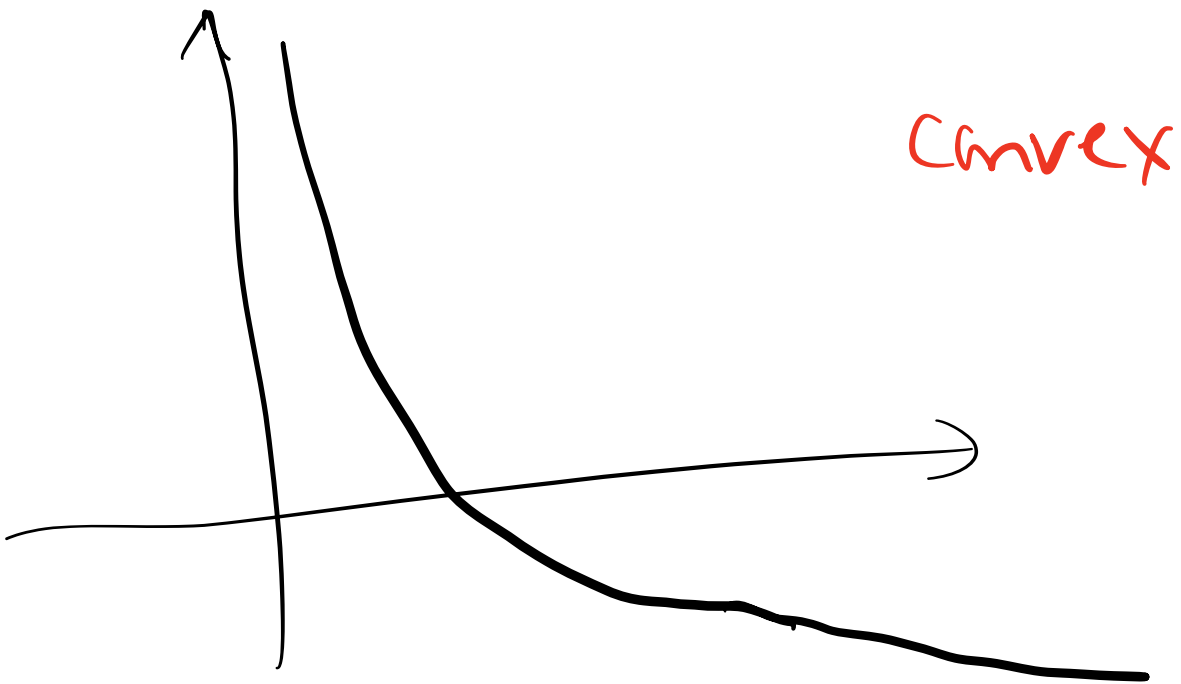
$$\boxed{KL(p, q) \geq 0}$$

$$KL(p, q) = \sum_i p_i \log\left(\frac{p_i}{q_i}\right)$$
$$= E\left[\log\left(\frac{p}{q}\right)\right]$$



concave

but what about $-\log$?



$$\begin{aligned}
 KL(p, q) &= E \left[\log \left(\frac{p}{q} \right) \right] \\
 &= E \left[-\log \left(\frac{q}{p} \right) \right]
 \end{aligned}$$

$$f = -\log$$

$$f(E(x)) \leq E(f(x))$$

$$-\log(E(x)) \leq E(-\log(x))$$

$$\begin{aligned}
\Rightarrow KL(p, q) &= E \left[-\log \left(\frac{q}{p} \right) \right] \\
&\geq -\log \left(E \left(\frac{q}{p} \right) \right) \\
&= -\log \left(\sum_i p_i \frac{q_i}{p_i} \right) \\
&= -\log \left(\sum_i q_i \right) \\
&= -\log(1) = 0.
\end{aligned}$$

$$KL(p, q) = 0 \Leftrightarrow p = q$$